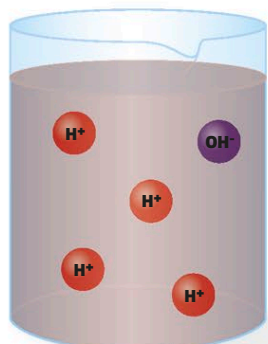
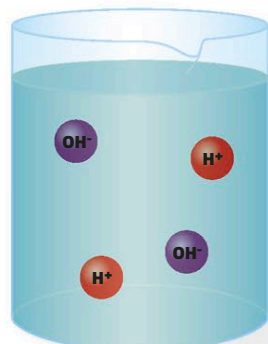


It's all about the ions

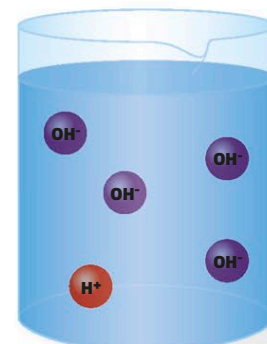
The difference between an acid and an alkali comes down to their proportion of positively charged particles called hydrogen ions (H^+). When an acidic compound is dissolved in water, it breaks up, releasing H^+ ions: it has an increased proportion of positively charged ions. When an alkaline compound dissolves in water it releases negatively charged particles called hydroxide ions (OH^-). Acids are called H^+ donors; alkalis are called H^+ acceptors.



Acid
There are more positively charged H^+ ions than negatively charged OH^- ions in an acid.



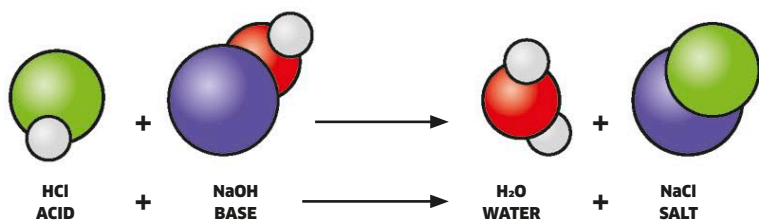
Neutral
A neutral solution contains equal numbers of positive H^+ and negative OH^- ions.



Base
There are more negatively charged OH^- ions than positively charged H^+ ions in an alkali.

Mixing acids and bases

The reaction between an acid and an alkali produces water and a salt. It is called a neutralization reaction. The H^+ ions in the acid react with the OH^- ions in the alkali, resulting in a substance that is neither acid nor alkali. Different acids and alkalis produce different salts when they react.



Neutralization formula

When hydrochloric acid (HCl) reacts with the alkali sodium hydroxide ($NaOH$), they produce a neutral solution that consists of water (H_2O) and a well-known salt—sodium chloride ($NaCl$), or table salt.

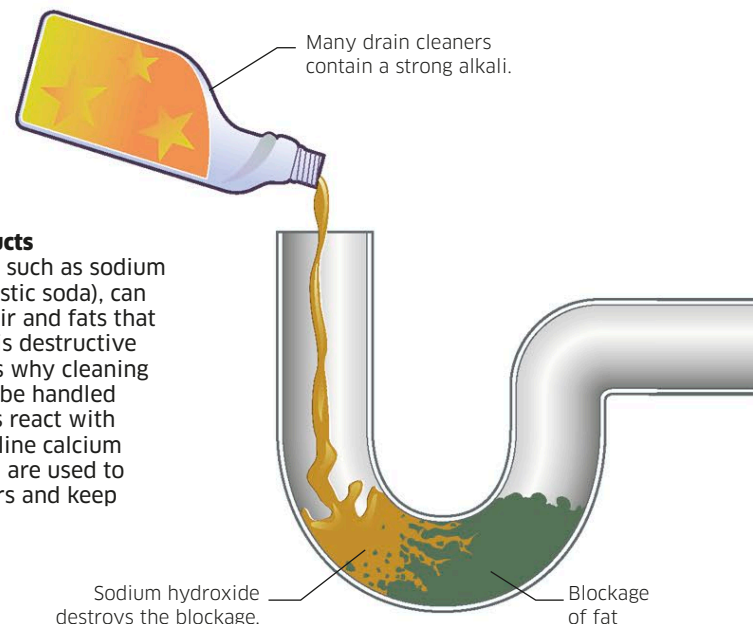


Acids and bases in agriculture

Farmers monitor soil pH levels carefully. Soils are naturally acidic or alkaline, and different crops prefer a higher or lower pH. Farmers can reduce the soil pH by adding certain fertilizers, or raise the soil pH with alkalis, such as lime (calcium hydroxide).

Kitchen chemistry

The kitchen is a great place to see acids and alkalis in action. Weak acids—found in lemon juice and vinegar—can preserve or improve the flavor of food. When baking, we use weak alkalis present in baking soda to help cakes rise. Strong acids and alkalis are key ingredients in a range of cleaning products. They are so powerful that protective gloves must be worn when using them.



Cleaning products

A strong alkali, such as sodium hydroxide (caustic soda), can break down hair and fats that clog drains. This destructive power explains why cleaning products must be handled carefully. Acids react with limescale (alkaline calcium carbonate) and are used to descale showers and keep taps shiny.

Bubbles made by carbon dioxide

Baking powder

Added to flour to help cakes rise, baking powder contains an acid and an alkali, which react together when a liquid and heat are added. The reaction produces bubbles of carbon dioxide that push the cake mixture upward.

